

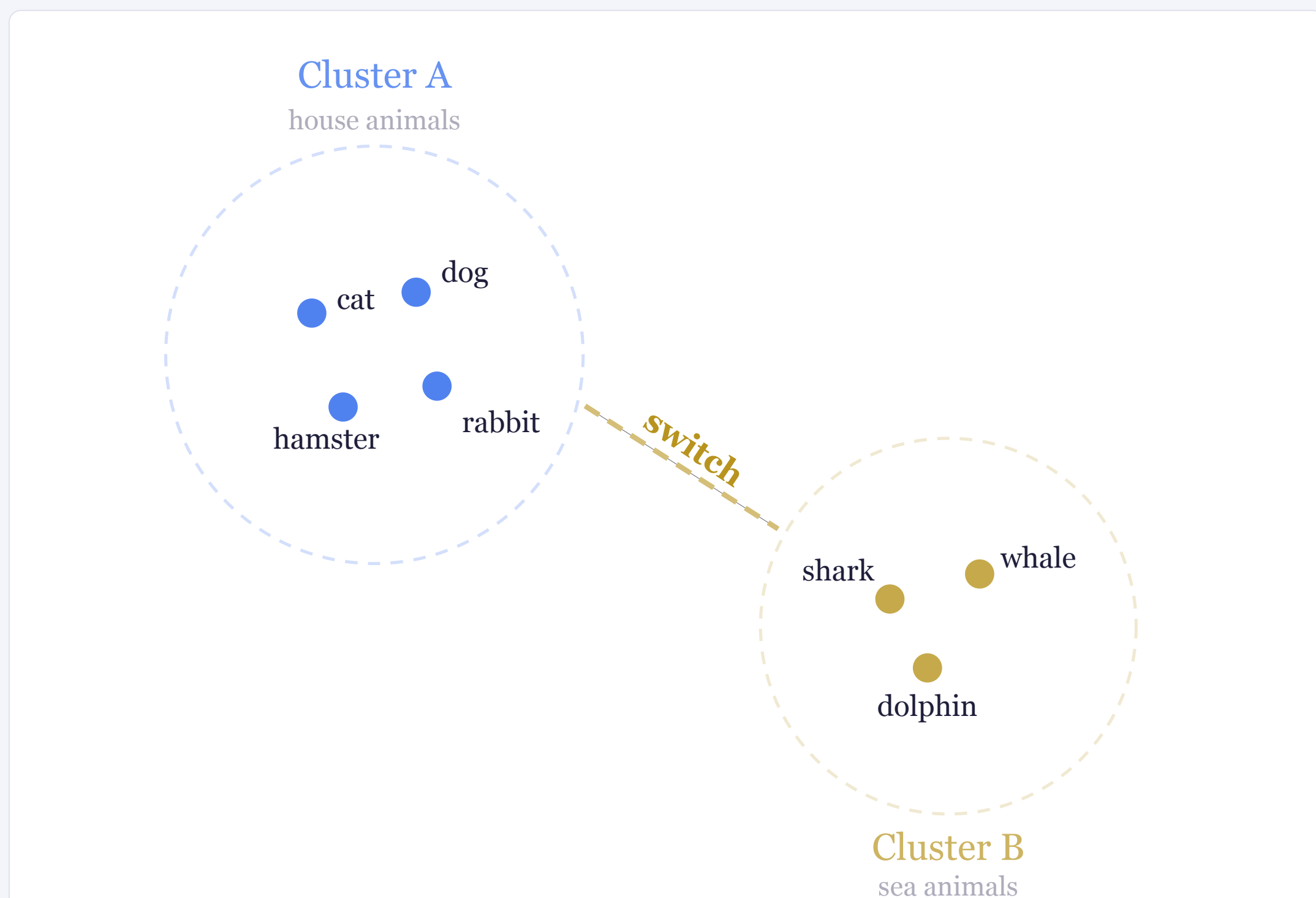
Characterizing human semantic navigation as trajectories in embedding space

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Semantic navigation: moving through concepts

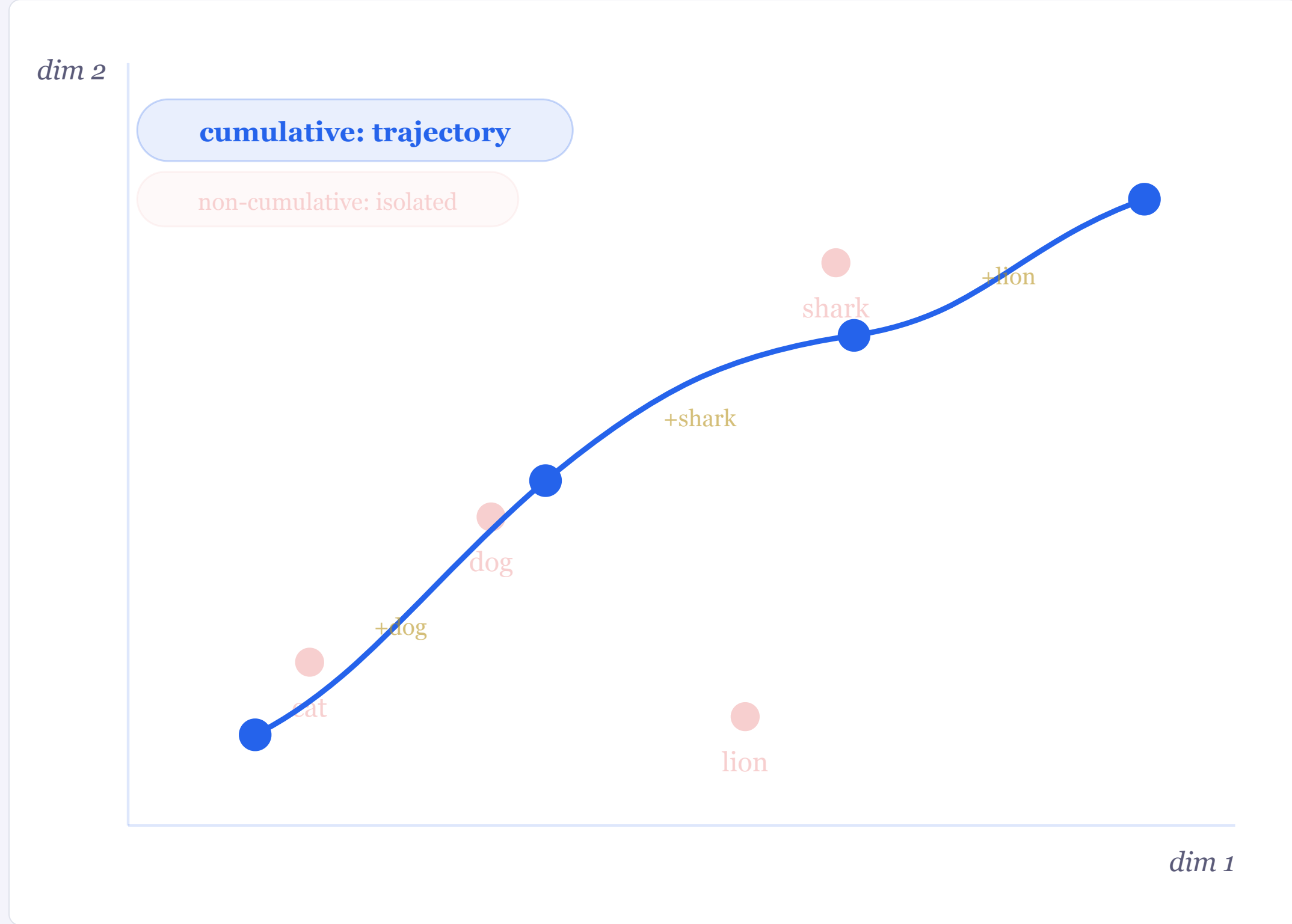


Semantic retrieval is often characterized as *clustering* and *switching*; traditional pipelines are *labor-intensive*, heterogeneous, and miss the *step-by-step granularity* of how meaning unfolds over time.

→ What if we framed semantic retrieval as a trajectory through geometric space?

Cumulative embeddings encode search history

Each step encodes the **full prefix**: embed "cat" → "cat dog" → "cat dog shark" ... capturing dependencies and working memory. Result: a trajectory per participant × concept pairs.



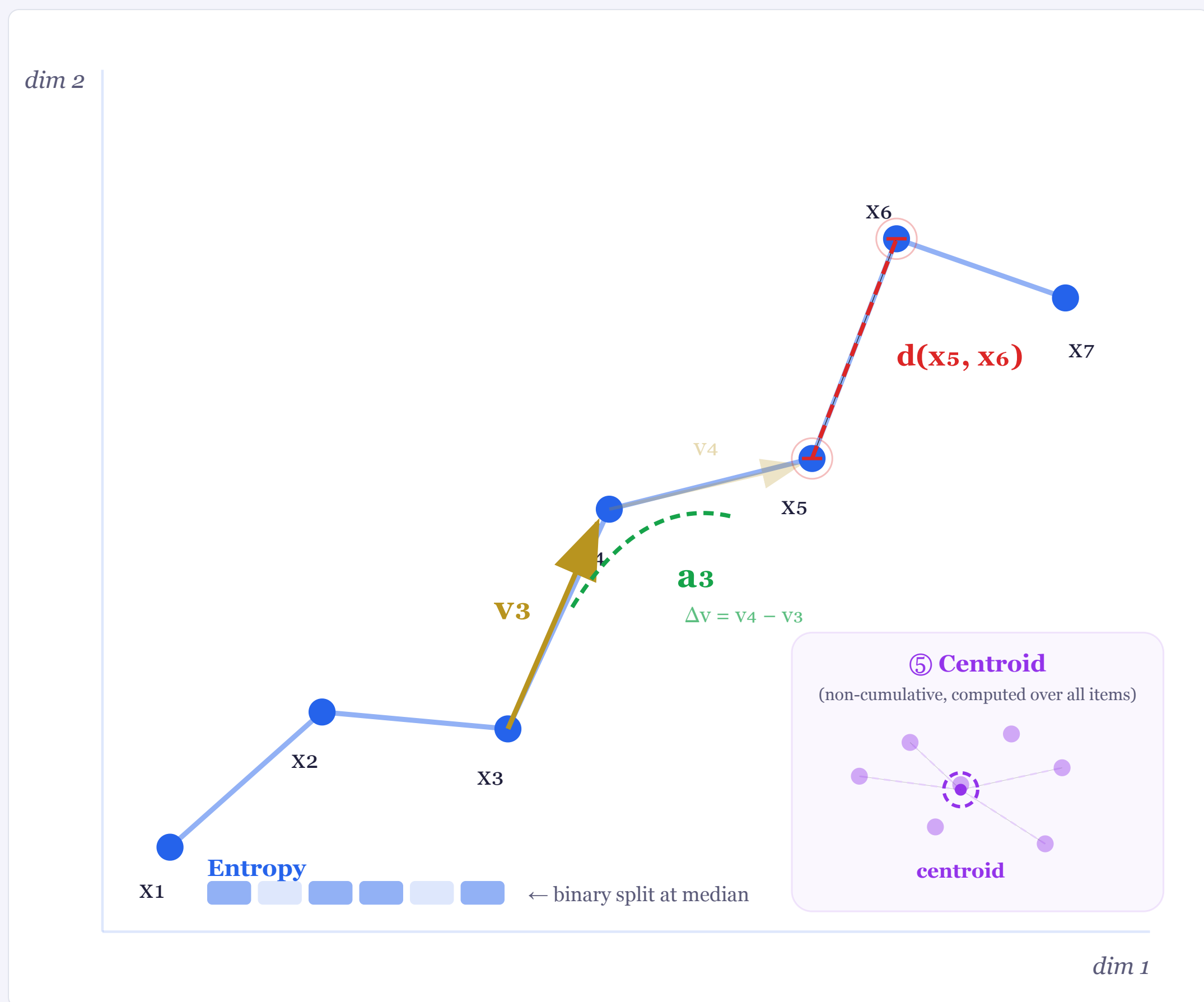
Four datasets, four languages, two tasks

Statistics: GLMMs with Tukey HSD post-hoc.

A Neurodegenerative	ES-CL · N=76 · Property Listing · HC / PD / bvFTD
B Swear Fluency	EN · N=274 · Verbal Fluency · Swear / Animals / Letters
C Cross-Linguistic PLT	IT (N=69) & DE (N=73) · 10 categories · Property Listing

Embedding models	Design choices
- OpenAI text-embedding-3-large - Google text-embedding-004 - Qwen3-Embedding-0.6B - fastText (baseline, non-cumul.)	- Causal and bidirectional attention - Cumulative vs. non-cum. comparison - ZCA-whitening tested for anisotropy

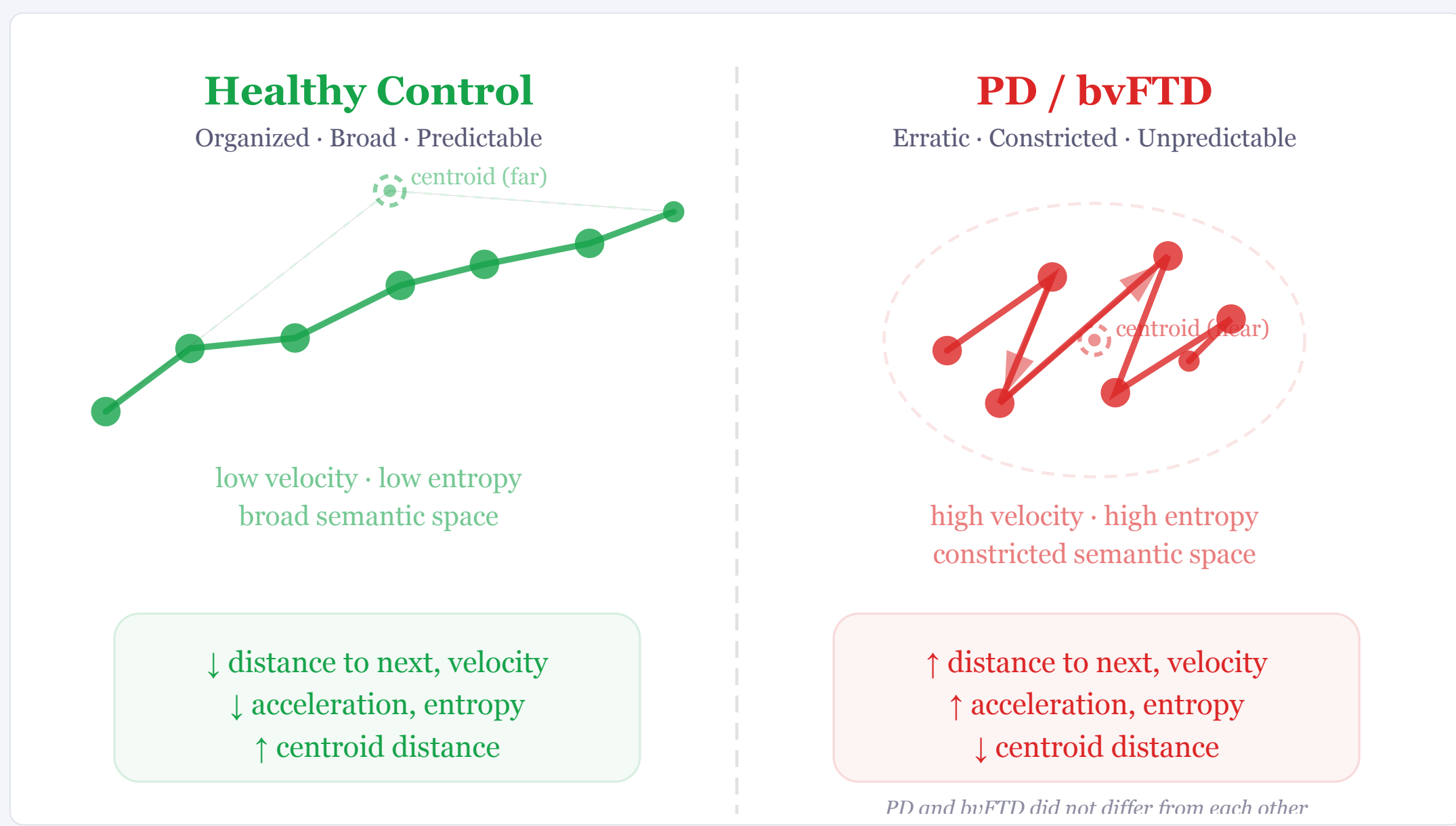
Five physics-inspired trajectory metrics



① Distance to next Cosine distance between consecutive points. <i>Semantic jump size</i> .	③ Acceleration $a_t = v_{t+1} - v_t$ Low → stable cluster. High → erratic switch.
② Velocity $v_t = x_{t+1} - x_t$ Direction + magnitude of each step.	④ Entropy Shannon entropy of median-split steps. <i>Predictability of the search</i> .

⑤ Distance to centroid Distance to mean position of all items. <i>Dispersion of the search</i> .
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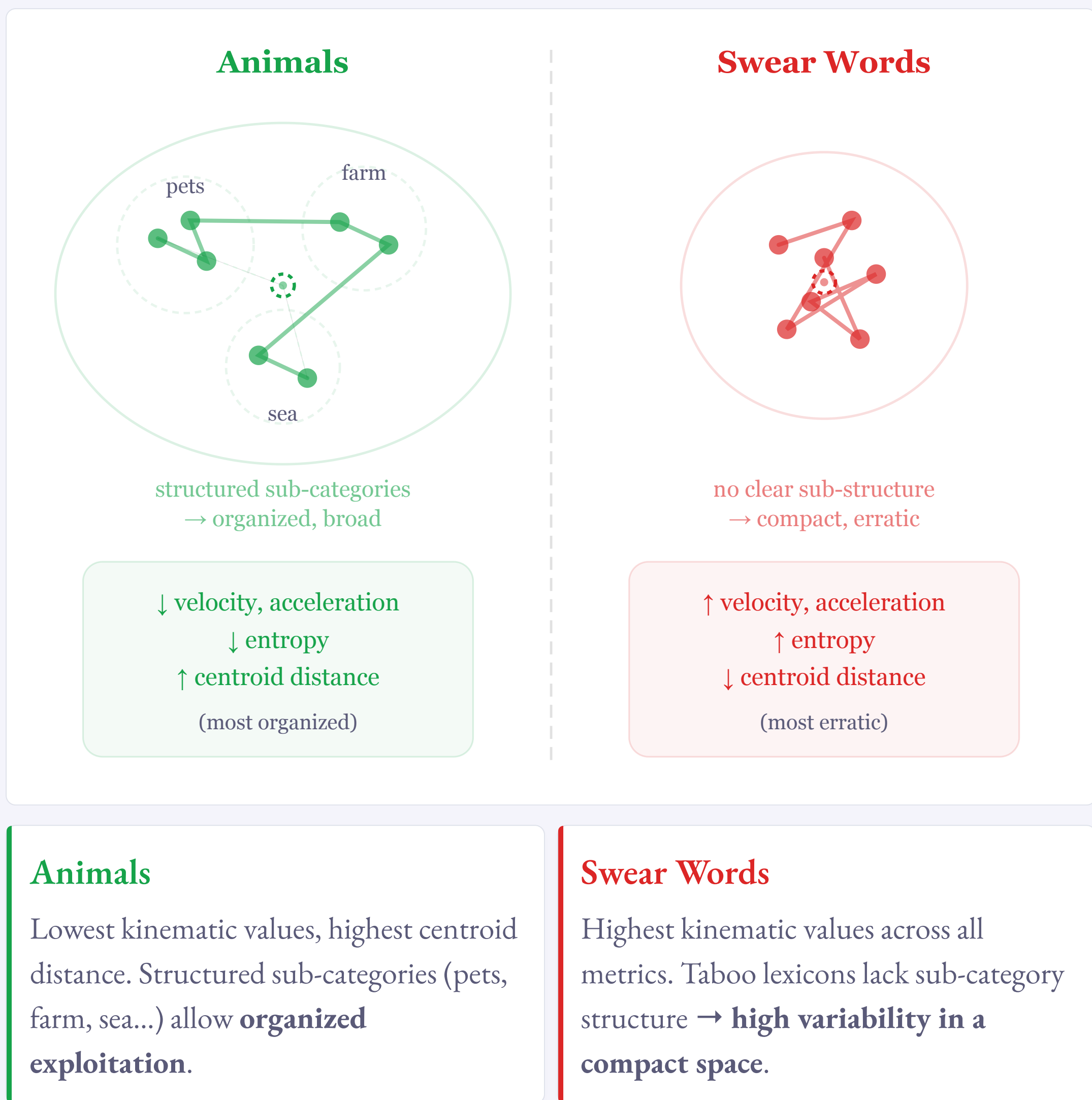
Patients show more erratic, constricted trajectories



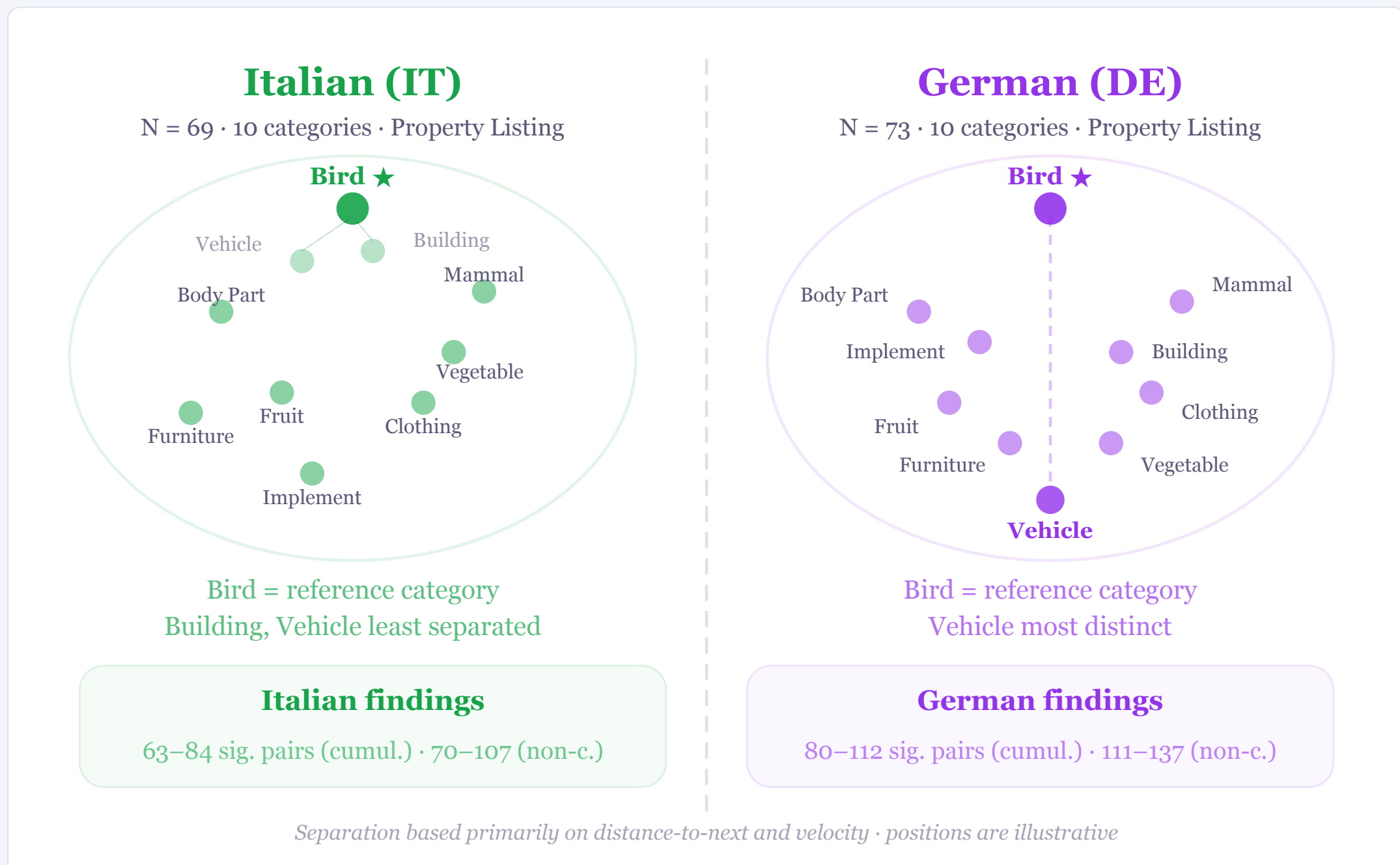
↑ Higher in patients Dist. to next — larger semantic jumps Velocity — erratic movement Acceleration — abrupt direction shifts Entropy — unpredictable search	↓ Lower in patients Distance to centroid — search confined to a tighter neighborhood despite being more volatile
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A kinematic signature of executive dysfunction — patients navigate semantic space with volatile, erratic steps yet remain confined to a narrow neighborhood.

Category structure shapes semantic kinematics



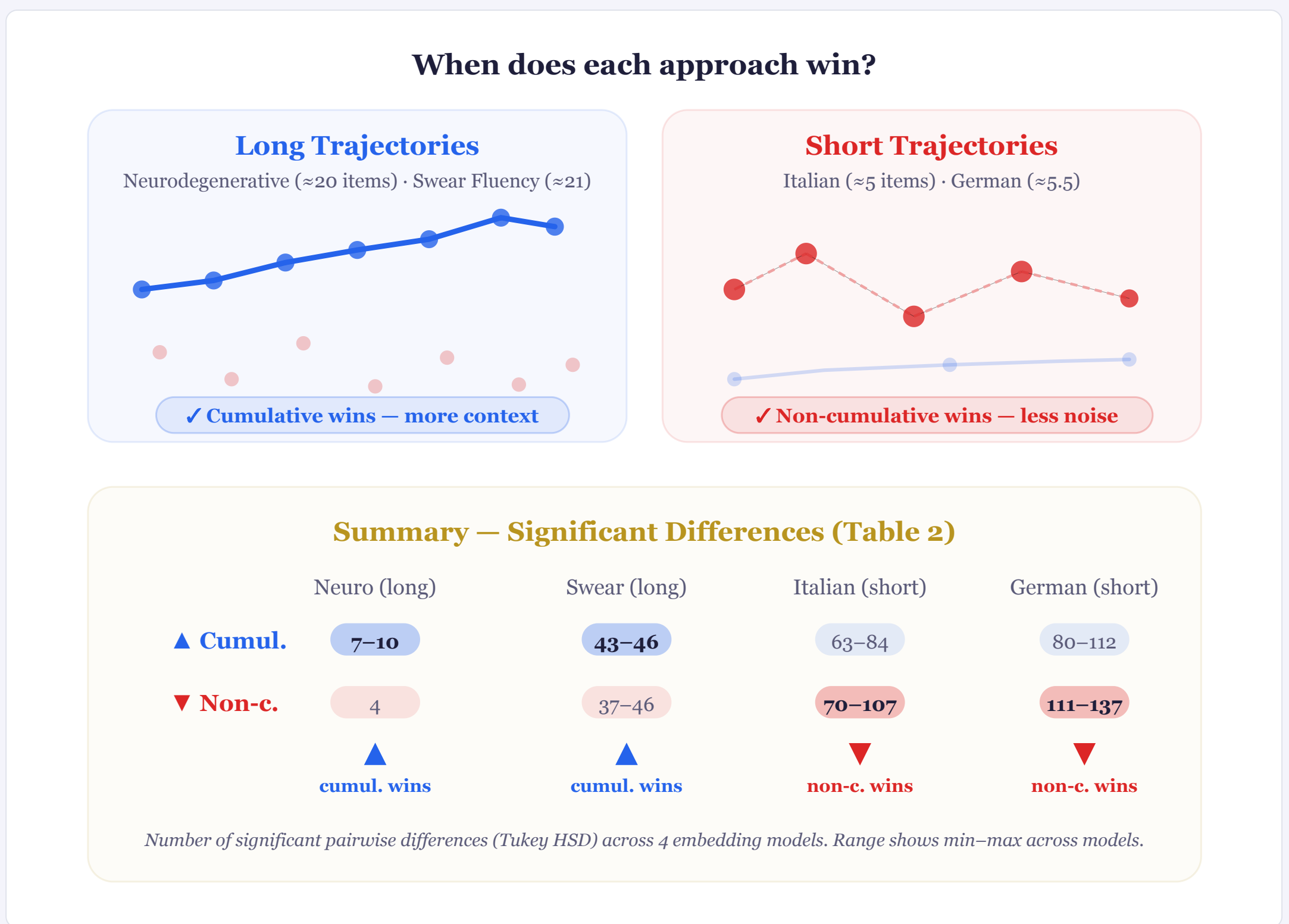
Language and culture shape semantic organization



Italian (IT) · N=69 10 categories · Property Listing Task. Bird is reference. Building and Vehicle least separated . Category effects <i>selective</i> , not uniform.	German (DE) · N=73 Same 10 categories · Same protocol. Vehicle most distinct from Bird. Bird highest velocity & acceleration. Fruit vs. tool-like categories show strongest centroid gaps.
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Same protocol, different category effects — cultural and linguistic structure shapes how concepts are organized in semantic space.

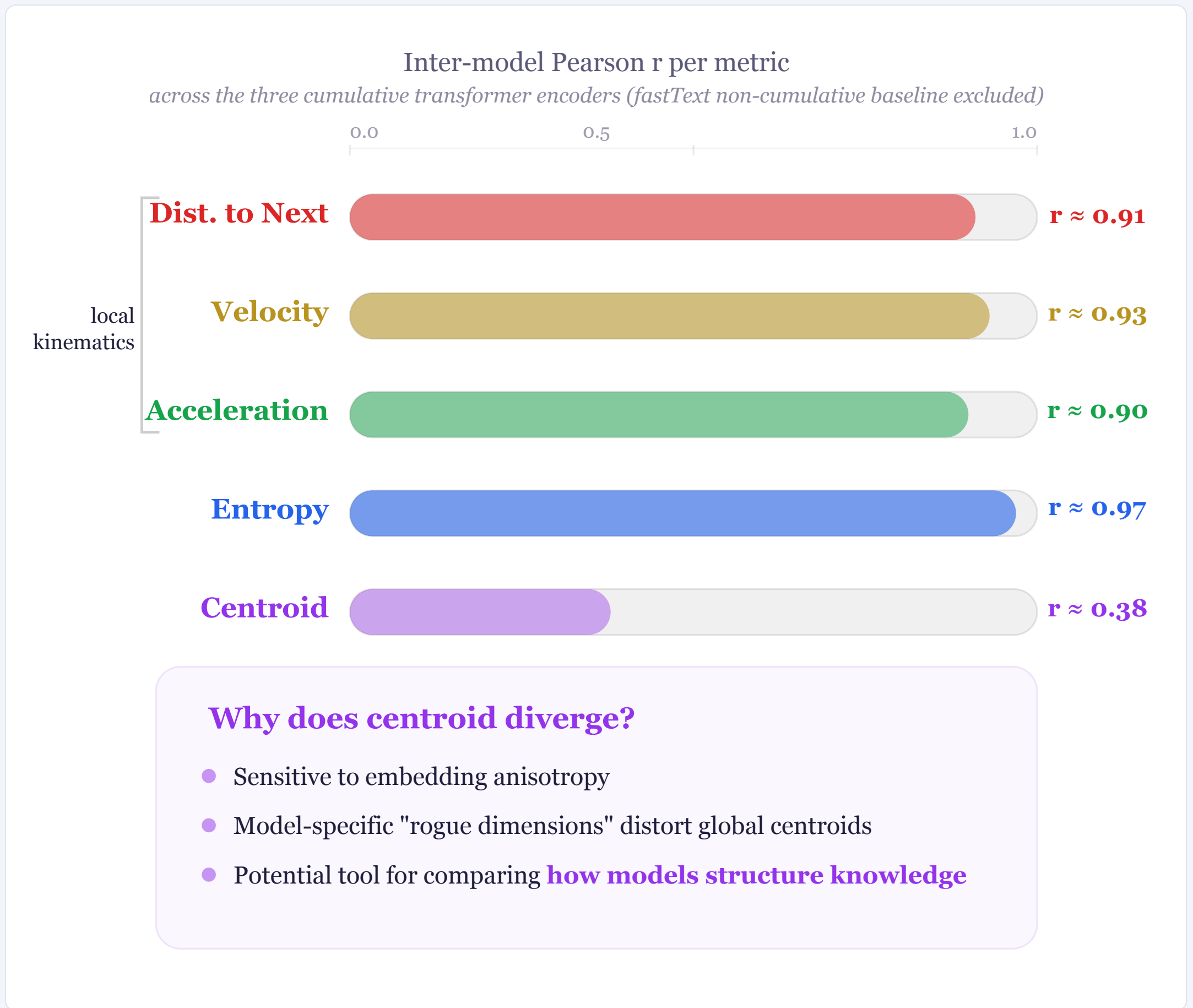
Cumulative wins for long sequences



Cumulative → long sequences	Non-cum. → short sequences
Longer trajectories provide rich context — more significant group differences with higher effect sizes across all metrics.	With ~5 items, there is too little context to accumulate. Point-to-point variation retains more discriminative signal.

Trajectory length determines which best exposes semantic structure.

Different models, convergent geometry



Towards a framework for semantic navigation

- Cumulative metrics build on clustering/switching, capturing fine-grained dynamics
- Clinical utility — kinematic signatures of neurocognitive dysfunction
- Discriminates categories & reveals language-specific organization
- Model-robust — local kinematics converge across embedding architectures

Future: temporal timestamps · more mathematically robust metrics



Paper



Code